The Impact of Remote Work on Women's Labor Market Share

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Abstract

This paper examines whether remote work can improve women's representation in the U.S. labor market. Using data from the Quarterly Workforce Indicators (QWI) and job postings, I estimate the causal effects of changes in remote work share on the female share among new hires and in the workforce. To identify these effects, I employ two strategies: a difference-in-differences (DID) framework augmented with double machine learning (DML), and an instrumental variable (IV) approach. The results show that transitioning jobs from fully on-site to arrangements that include remote working days would raise the female share of new hires by about 10 percentage points and the overall female workforce share by about 6 percentage points. These gains are persistent over time and particularly pronounced in male-dominated industries. The findings highlight remote work as a promising mechanism for advancing gender equality in the labor market and increasing female labor force participation.

Keywords: Remote work, Women's labor market representation, Gender equality

JEL: J21, J63, J71

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1 Introduction

Women have historically been underrepresented in the labor market. In 1970, females accounted for less than 40% of the total employed workforce in the United States. Over the following decades, female labor force participation increased rapidly, and by 2000 women represented 46.5% of the workforce. Since then, however, progress has largely stalled: by 2019, the female share had reached only 47.1%. This persistent under-representation is particularly striking given that women outnumber men in the overall population. As shown in Figure 1, the employment-to-population ratio gap between men and women has gradually narrowed but remains around 10 percentage points. Across the 19 industry sectors shown in Figure 2,[1] 12 have a female workforce share below 50%.

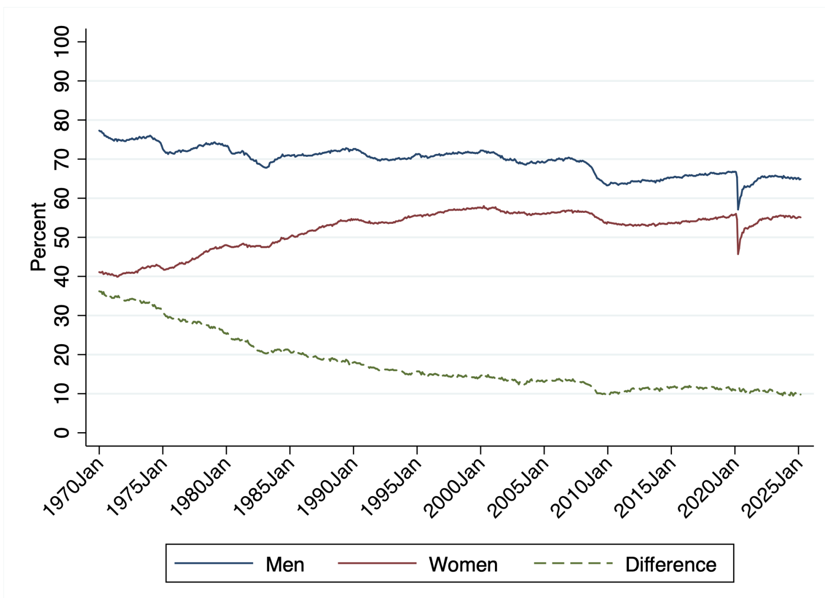

Figure 1: Employment-to-population ratio by gender, U.S. Jan 1970–Feb 2025

Why does this gap persist? Explanations based on human capital differences have become untenable over the course of the twenty-first century. First, technological progress has steadily reduced the prevalence of physically demanding jobs, meaning that biological differences in physical strength no longer constitute a major barrier to women's labor force

[1] Data from Quarterly Workforce Indicators (QWI), excluding public administration due to data availability.

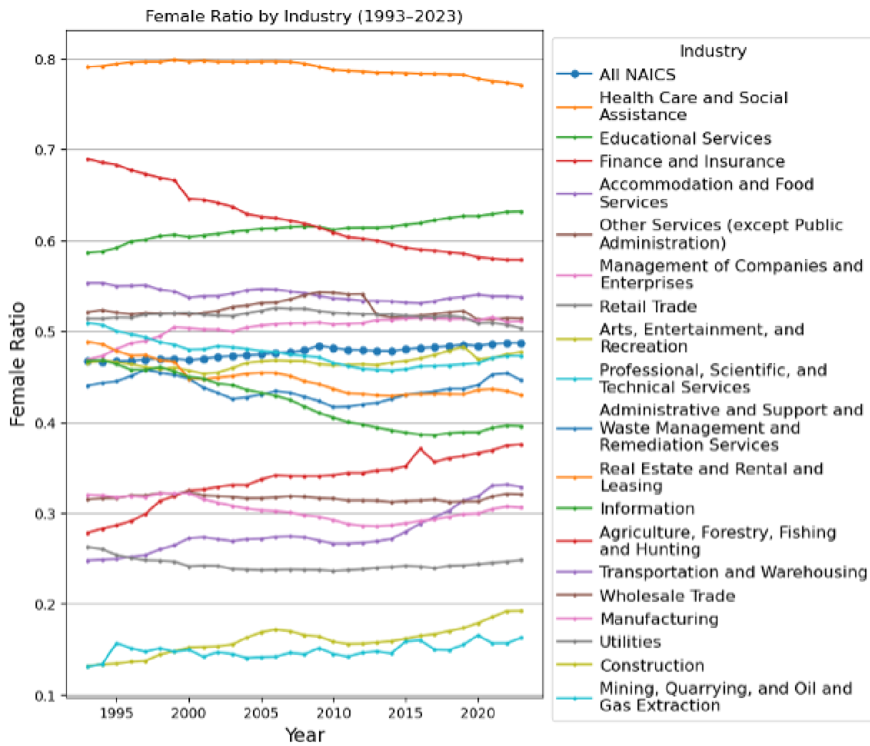

Figure 2: Female share by industry, U.S. 1993-2023

participation. Second, women have surpassed men in educational attainment, eliminating the disadvantage of lower schooling that characterized much of the twentieth century.[2] These changes suggest that other barriers beyond human capital play a central role in sustaining gender inequality in labor market outcomes.

First, caregiving responsibilities remain a key constraint on women's labor market outcomes. Domestic responsibilities, particularly child care, continue to fall disproportionately on women, making rigid work arrangements in terms of hours and location a significant barrier. Second, women's career choices are constrained by household dynamics, most notably the "two-body problem." Evidence shows that in dual-earner households, family location decisions often prioritize men's careers, limiting women's geographic mobility and professional advancement (Bielby and Bielby, 1992; Cooke, 2003; Nivalainen, 2005; Swain and Garasky, 2007; Sorenson and Dahl, 2016). These patterns reflect the persistence of traditional gender

[2]According to the 2020 American Community Survey (ACS), among individuals aged 25–54, 42% of women hold a college degree compared to 35% of men.

role ideologies, which continue to restrict women's labor supply.

On the one hand, these barriers constrain women's labor supply, and employers' perceptions of the risks associated with hiring female workers often lead to discrimination, which helps explain women's disadvantaged position in the labor market. On the other hand, they underscore the importance of flexibility as a potential mechanism for narrowing gender gaps. Even before 2020, when remote work was still rare, research had already documented women's stronger preference for job flexibility (Flabbi and Moro, 2012; Mas and Pallais, 2017; Wiswall and Zafar, 2017). Claudia Goldin, recipient of the 2023 Nobel Prize in Economics, emphasized the crucial role of job flexibility in advancing gender equality in the labor market (Goldin, 2014). Although the theoretical foundation for the impact of remote work on women's labor supply decisions is well established, the limited availability of remote jobs prior to 2020 made it difficult to establish direct causal evidence. Stimulated by the COVID-19 pandemic, remote work has become substantially more prevalent. By 2024, around 10 percent of job postings in the United States contained a remote component (Hansen et al., 2023). Moreover, the expansion of remote work has varied considerably across industries and locations.[3] The growth and the accompanying cross-sectional variation of remote work provide a valuable research context for studying the role of job flexibility in women's labor market outcomes.

My study focuses on how remote work affects women's representation in the labor market. Specifically, I examine whether shifting jobs from a traditional five-day, fully on-site schedule to arrangements that incorporate remote working days increases the share of women among new hires and in the overall workforce.

In this study, I use data from the Quarterly Workforce Indicators (QWI) to measure female representation in the workplace and data from Hansen et al. (2023) to capture the prevalence of remote work. My analysis is conducted at both the county–industry and county levels, depending on the specification. The identification strategy relies on a difference-

[3] Detailed trends and data are available at https://wfhmap.com/.

in-differences (DID) framework. First, I estimate a two-period DID at the county–by-industry level, using the female shares in new hires and in the workforce as the outcomes, and changes in remote work share as the treatment. To address concerns that unobserved county–industry-specific shocks related to local or industry characteristics other than remote work may also affect the outcomes, I include a set of covariates and apply double machine learning (DML) techniques, which allow for flexible, non-linear relationships between covariates and outcomes. Second, as a complementary robustness analysis, I instrument changes in the county-level remote work share using counties' differential exposure to industry-level shifts in remote work adoption. This instrumental variable (IV) approach mitigates concerns about unobserved selection into remote work. Both identification strategies leverage variation in industry-level remote work adoption, which is primarily determined by task characteristics and can plausibly be treated as exogenous.

My study finds that increasing remote work raises female representation in the workplace. Specifically, shifting all jobs in a workplace from fully in-person to arrangements that include remote working days would increase the female share of new hires by about 10 percentage points and the overall female workforce share by about 6 percentage points. This effect is persistent and remains significant even in male-dominated industries. In addition, I examine the sources of this increase and find that it is driven by a combination of female job switchers and new entrants to the labor market.

This paper provides empirical evidence for the causal effect of remote work on women's share in the labor market, showing that traditional job structures with fixed hours and locations constitute significant barriers to female labor force participation. It contributes to the literature on remote work and gender inequality by offering novel evidence that remote work can advance women's representation and has the potential to reduce gender segregation in the labor market. In addition, the findings carry practical implications for policymakers and organizations seeking to promote workplace diversity and equity. Finally, in the context of population aging and declining fertility rates, remote work may help mitigate labor shortages by mobilizing greater female labor force participation.

The paper is structured as follows. Section 2 reviews the related literature. Section 3 describes the data, measurement, and descriptive statistics. Sections 4 and 5 present the empirical strategy and the main results. Section 6 explores potential mechanisms. Sections 7 shows the robustness check results, and Section 8 discusses the implications and limitations. Finally, Section 9 concludes the paper.

2 Related Literature

The first branch of literature relevant to my study focuses on women's preference for job flexibility, providing a theoretical foundation for assessing how remote work may shape female labor market outcomes. These preferences were well established before the COVID-19 pandemic, at a time when remote work opportunities were still limited. Flabbi and Moro (2012), using CPS data, estimate that more than one-third of college-educated women positively value flexible jobs, with the value ranging from 1 to 10 cents per hour. Mas and Pallais (2017) use an experiment in the hiring process of a national call center to estimate the willingness-to-pay (WTP) distribution for alternative work arrangements. They find that women place a higher value than men on working from home and on avoiding irregular work schedules. Women appear willing to pay more than twice as much as men for the ability to work from home. Likewise, they are willing to pay nearly twice as much as men to avoid employer discretion in scheduling. Notably, women with young children exhibit an even higher WTP for working from home than other women. Similarly, Wiswall and Zafar (2017), using a survey of NYU undergraduates, find that women exhibit a much stronger preference for workplace hours flexibility, with an implied WTP of 7.3% of annual salary compared to 1.1% for men. These results align with evidence that women are more likely to be employed in jobs that offer greater flexibility (Goldin and Katz, 2011; Wasserman, 2022).

With the widespread adoption of remote work after the pandemic, recent studies have revisited flexibility preferences using broader samples and geographic contexts. Richards et al. (2024), analyzing surveys in Ireland, show that women express a stronger desire to work from home post-pandemic. Similarly, Artz et al. (2025), using data from the Survey of Working Arrangements and Attitudes in the United States, find that women consistently report a stronger preference for WFH than men. Together, these studies demonstrate both pre- and post-pandemic evidence that women value workplace flexibility more highly, suggesting that remote work may be particularly relevant for female labor supply decisions.

Research on gender segregation in the labor market is also relevant to my study, as this literature documents both the patterns and dynamics of segregation across industries and occupations and highlights the role of women's choices in shaping these outcomes. Women and men remain unevenly distributed across industries and occupations (Blau et al., 2012; Blau and Kahn, 2017; Hegewisch and Williams-Baron, 2017; Cortés and Pan, 2023). A supply-side explanation emphasizes that women disproportionately sort into lower-paying but more flexible jobs in order to balance work and family responsibilities. Barbulescu and Bidwell (2013), examining the job search behavior of MBA students, find that women are less likely than men to apply for finance and consulting jobs and more likely to apply for general management positions, partly because of their preferences for work–life balance. Hegewisch and Williams-Baron (2017) further highlight the role of women's choices in shaping labor market outcomes by showing that states with the most extensive work–family supports have the lowest gender wage gaps, while those with the least supports have the highest. This evidence reinforces the view that women's preferences contribute to sustaining occupational segregation.

Finally, a small but growing body of recent work examines how remote work may directly alter female labor supply, which motivates my research. Tito (2024) provides evidence that remote work enhances women's labor force attachment by reducing the probability of labor force exit. Hsu and Tambe (2025) show that switching a job posting from on-site to remote increases the share of female applicants by about 15%. These studies highlight the potential of remote work to change female labor supply, raising the critical question that my study addresses: whether these shifts ultimately translate into greater female representation in the labor market.

3 Data, Measurements, and Descriptive Statistics

The data of study mainly come from two sources: the Quarterly Workforce Indicators (QWI) and job posting data from Hansen et al. (2023).

Female employment data used in this study are drawn from the Quarterly Workforce Indicators (QWI), a set of economic indicators that include employment, job creation and destruction, wages, hires, and other measures of employment flows. These indicators are reported by detailed firm characteristics and worker demographics. I extract employment, hiring, and separation data by gender for each county and 2-digit NAICS industry from the QWI. From these data, I calculate the female share of total employees and new hires, as well as separation rates for both genders. In addition, I use the QWI to construct county–industry characteristics that serve as covariates in the double machine learning model, including the distributions of firm size, firm age, worker age, gender, education, race, and ethnicity.

To measure the share of remote jobs across industries and counties, I use data from Hansen et al. (2023), *"Remote Work across Jobs, Companies, and Space."* The authors employ a large language model (LLM) to analyze job postings and classify whether they offer remote work at least one day per week. Data used in my study monthly is provided by the research team at the industry–occupation–county level.

I define a remote job as one that offers remote work at least one day per week. The remote share for group g in period t is calculated as the proportion of job postings classified as remote within the group. Specifically, the measure is computed as follows:

$$\text{RemoteShare}_{g,t} = \frac{\#\text{Remote job postings in group } g \text{ during period } t}{\#\text{Total job postings in group } g \text{ during period } t}, \tag{1}$$

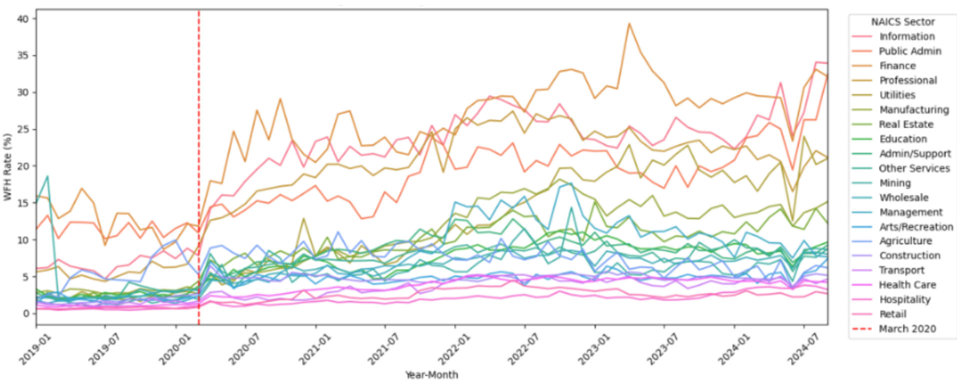

Figure 3: Remote job rate by industry, U.S. Jan 2019–Sep 2024

where g denotes either a county or an industry-county pair. This measure captures the share of jobs within a workplace—defined as either a county or an industry within a county—that are not traditional fully on-site positions but instead allow some degree of flexibility, including both hybrid and fully remote arrangements.

Figure 3 presents the evolution of remote job rates across industries over time. The share of remote jobs in the U.S. began to rise sharply in the second quarter of 2020, following the onset of the COVID-19 pandemic. Both the level and the change in remote job shares exhibit substantial variation across industries. Highly remote industries—such as Information, Public Administration, Finance, and Professional Services—experienced increases of 10 to 20 percentage points, whereas low-remote industries, including Retail, Hospitality, and Health Care, saw changes of less than 3 percentage points.

Compared to industry-level variation, county-level differences in remote work shares are smaller but still notable. Figure 4 presents maps that illustrate the spatial distribution of remote job shares and their changes. On average, the remote job share across counties increased from 2.8% in 2019 to 5.8% in 2021. During the same period, the variation across counties also widened, with the standard deviation rising from 3.7 percentage points to 5.3 percentage points. While some counties saw minimal changes in remote job share, others experienced substantial shifts, with the maximum increase reaching 48 percentage points.

I measure the change in remote work share for each county-industry group by comparing

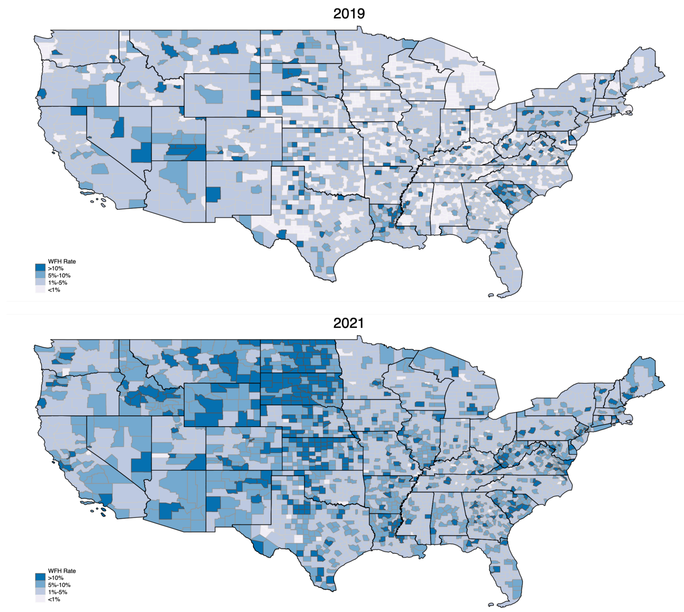

Figure 4: Remote job rate by county, U.S. 2019 v.s. 2021

the value in 2019 to the early post-pandemic period (April 2020 to December 2021). Specifically, I compute the difference in remote share—defined in Equation (1)—between these two periods using Equation (2). The change in remote share can arise from two sources: existing jobs shifting from on-site to remote arrangements, or the creation of new remote jobs. Since I calculate the change using periods immediately before and after the pandemic, the difference primarily reflects adjustments in existing jobs rather than the emergence of entirely new positions.

Figure 5 shows the distribution of these changes. On average, the remote share increased by 4.1 percentage points. About 33% of county-industry groups experienced a change of less than 1 percentage point, while 10.6% of groups saw an increase greater than 10 percentage points.

$$\Delta Remote\, Share_{ic} = Remote\, Share_{ic,04/2020-12/2021} - Remote\, Share_{ic,2019}. \tag{2}$$

Figure 6 presents the female share and the remote work share for each industry. The female share is based on data from Q4 2019, while the remote share reflects the average

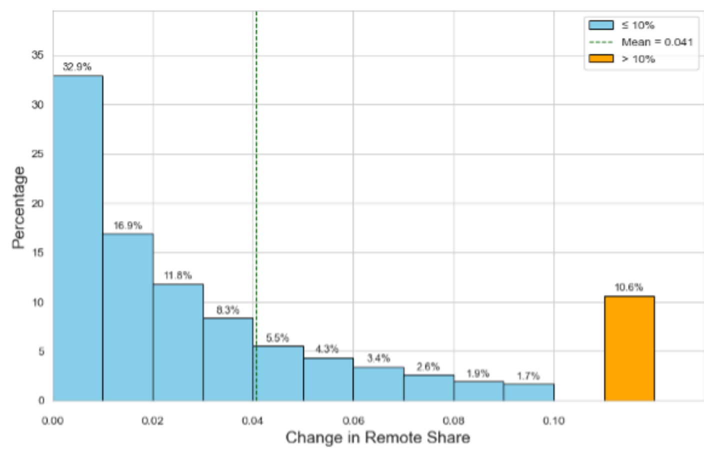

Figure 5: Change in remote job share by county-industry group, U.S. from year 2019 to Apr 2020-Dec 2021

remote work share in the post-pandemic period (from April 2020 to September 2024). The graph illustrates the relationship between an industry's remote work intensity and its gender composition. Among the 19 industries, 12 are male-dominated (with more than 50% male workers). Of the 6 high-remote industries—defined as those with an average post-pandemic remote share above 10%—only one has a female majority (more than 50% of workers are women). This pattern reflects the broader gender imbalance in the labor market, particularly within high-remote industries.

4 Identification Strategy

This section outlines the identification strategies used in the paper to estimate the causal effects of remote work on female representation in the labor market, focusing on the female share of new hires, the female separation rate, and the overall female workforce share. The analysis is based on a difference-in-differences (DID) framework. To address potential endogeneity concerns, I employ double machine learning (DML) and instrumental variable (IV) methods.

In my analysis, the pre-period is defined as 2019, the year prior to the COVID-19 pandemic when remote jobs were not yet prevalent. The post-period is defined as April 2020 to

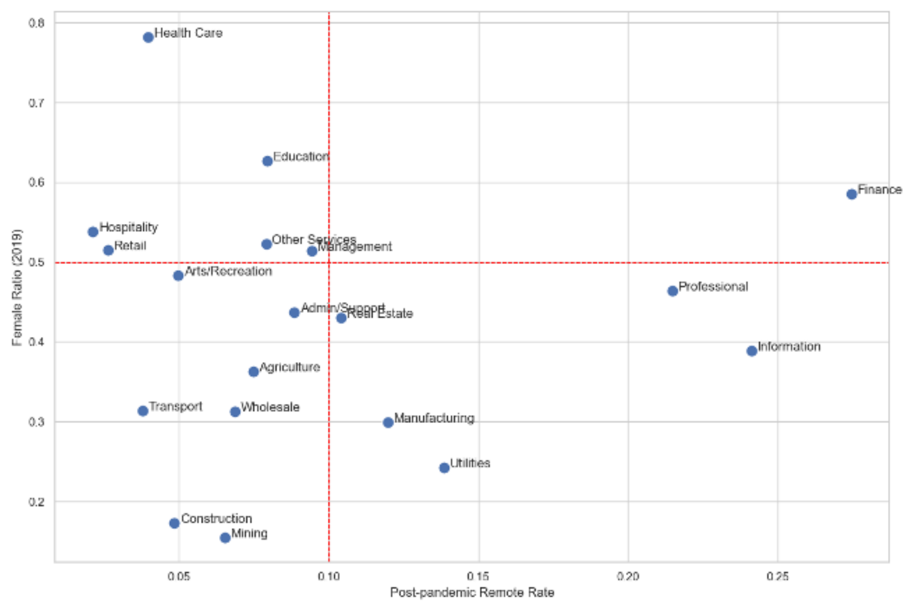

Figure 6: Industry female share in 2019 and change in remote job share

December 2021, capturing the initial rise of remote work. While most of the analysis focuses on this period, when studying the effects on female workforce share, I extend the post-period to 2022 Q4, 2023 Q4, or 2024 Q3 to examine longer-term effects.

The canonical model of the analysis is a two-period Difference-in-Differences (DID) model with continuous treatment:

$$Y_{g,t} = \alpha + \beta^{DD} Post_t \times \Delta Remote\, Share_g + \gamma_g + \sigma_t + \epsilon_{g,t}, \tag{3}$$

where g denotes the group of the observation. In the baseline analysis, I define each county-by-industry combination as a group.[4] $Y_{g,t}$ denotes the outcome observed for group g in the last quarter of the pre- and post-periods, specifically 2019 Q4 and 2021 Q4. In analyses where the post-period is extended, I instead use outcomes from 2022 Q4, 2023 Q4, or 2024 Q3. The treatment variable is defined as the change in the remote job share for group g, calculated as the difference in remote share between the pre- and post-periods. The regression model includes group fixed effects (γ_g) and time fixed effects (σ_t).

[4]In the later specification of the IV model, I alternatively define groups at the county level.

The regressions are weighted by the total employment of each group in 2019, giving more weight to larger groups. This reflects the idea that smaller groups are more sensitive to changes in worker counts. For example, an increase of 10 female workers could substantially alter the female share in a small group, while the same numerical change would have a much smaller impact in a larger group. By weighting more heavily toward larger groups, the analysis places greater emphasis on changes that are less likely to be driven by small-sample fluctuations.

Since the model includes only two time points, it can be rewritten as:

$$\Delta Y_g = \alpha + \beta^{DD}\, \Delta \text{Remote Share}_g + \epsilon_g. \tag{4}$$

Equation (4) presents the difference-in-differences (DID) specification in a concise form. It captures the relationship between changes in the share of remote work and changes in the outcome variable across groups. The coefficient β^{DD} can be interpreted as a causal effect under the parallel trends assumption, which posits that, in the absence of differential changes in remote work share, outcomes across groups would have evolved along similar trends. Equivalently, this implies that $\Delta \text{Remote Share}_g$ in Equation (4) is assumed to be exogenous with respect to other factors affecting ΔY_g.

A traditional way to test the parallel trends assumption is to examine the presence of differential pre-trends across groups. However, in the context of this study, even if no differential pre-trends are observed, the parallel trends assumption remains a concern. Beyond changes in remote work, groups may have experienced different policy and economic shocks during the pandemic, which could also affect labor market outcomes. For example, county-level changes in the remote share ($\Delta \text{Remote Share}_c$) may reflect regional shocks. Heterogeneous exposure to the pandemic and differences in local policy responses could jointly influence both the extent of remote work adoption and employment outcomes, potentially introducing bias.

To credibly identify the causal effects of remote work, I employ two methods: Double Machine Learning (DML) and Instrumental Variable (IV) approaches.

4.1 Double Machine Learning (DML) Method

The variation in the treatment variable, $\Delta\text{Remote Share}_G$—defined as the change in remote work share—can be decomposed into three components: (i) county-wide variation, (ii) industry-wide variation, and (iii) idiosyncratic variation at the county-industry level. To credibly identify a causal effect, it is essential that the variation used in estimation be exogenous. Among the three types of variation, industry-level variation is the most exogenous, as it is largely determined by the characteristics of jobs in the industry, which are predetermined and unrelated to shocks during the pandemic. County-level variation is closely related to heterogeneous exposure to COVID-19 and local policies, which also affected outcomes. Idiosyncratic variation in remote work at the county-industry level is related to the characteristics of the groups, which may also influence their labor market activities.

To address endogeneity concerns, I control for potential confounders in Equation (4). To accommodate high-dimensional covariates and allow for nonlinear relationships between covariates and both the outcomes and the treatment, I implement a double machine learning (DML) estimator developed by Chernozhukov et al. (2018). Specifically, I control for county fixed effects and county-by-industry group characteristics, assuming that these factors affect both the outcomes and the treatment. I apply the following partially linear model:

$$\begin{aligned} \Delta Y_g = \beta^{DML}\Delta\text{Remote Share}_g + g_0(X_g) + \varepsilon_g, &\quad E[\varepsilon_g|\Delta\text{Remote Share}_g, X_g] = 0, \\ \Delta\text{Remote Share}_g = m_0(X_g) + \zeta_g, &\quad E[\zeta_g|X_g] = 0, \end{aligned} \tag{5}$$

where X_g is a high-dimensional vector of confounders, including county fixed effects, the proportions of workers by firm size, firm age, education, race, and ethnicity groups, as well as the proportions of female workers by age group.

Naive inference which pulg-in machine learning prediction $\hat{g}_0(X_g)$ in to Equation (5) to estimate the causal parameter is generally invalid. The use of machine learning methods introduces a bias due to regularization. To overcome regularization bias, DML takes the following steps, which is called orthogonalization:

1. Estimation of Conditional Expectation Functions (CEF): estimate the conditional expectation functions, $m_0(X_g) = E[\Delta\text{Remote Share}_g \mid X_g]$ and $l_0(X_g) = E[\Delta Y_g \mid X_g]$, through a machine learning algorithm, random forest.

2. Obtain residuals: compute $\hat{v}_g = \Delta Y_g - \hat{l}_0(X_g)$ and $\hat{u}_g = \Delta\text{Remote Share}_g - \hat{m}_0(X_g)$.

3. Regression on Residuals: regress $\hat{v}_g$ on $\hat{u}_g$ to obtain $\hat{\beta}^{DML}$.

To mitigate bias from overfitting, I implement 5-fold cross-fitting when estimating $\hat{\beta}^{DML}$. Specifically, the sample is randomly partitioned into five equally sized folds. In each iteration, four folds are used as the training sample and the remaining fold serves as the hold-out sample. I train the random forest learners m_0 and l_0 on the training sample and then generate out-of-sample predictions for the hold-out sample to obtain residuals. This procedure is repeated across all five folds so that each observation is used exactly once for out-of-sample prediction.

The identification of the causal estimator relies on the Conditional Independence Assumption (CIA), which requires that, conditional on county-by-industry group characteristics and county fixed effects, the change in remote share is unrelated to other factors that affect the outcomes. Under the DID framework, this implies that groups with the same characteristics within the same county would have followed parallel trends in the absence of variation in the change in remote share.

The double machine learning method relaxes the parallel trends assumption in Equation (3) through addressing potential endogeneity from observed characteristics of county-by-industry groups. To validate the method and account for unobserved selection, I implement an instrumental variable approach.

4.2 Instrumental Variable (IV) Method

To address these remaining concerns of unobservable selections, I implement an instrumental variables strategy using a Bartik-style instrument. The analysis under the IV metthod is conducted on the county level. This approach leverages national trends in remote work adoption across industries, combined with the pre-pandemic industry composition of each county, to isolate plausibly exogenous variation in the change in remote work exposure at the local level.

To address remaining concerns about unobserved selection, I implement an instrumental variables strategy using a Bartik-style instrument. The IV analysis is conducted at the county level. This approach leverages national trends in remote work adoption across industries, combined with each county's pre-pandemic industry composition, to isolate plausibly exogenous variation in the change in remote work driven by differences in remote job exposure.

I exploit county-level variation in the change in remote work share, $\Delta\text{Remote Share}_c$, and instrument it using a Bartik-style instrument:

$$Z_c = \sum_{i \in I} l_{i,c}^{2019} \cdot \Delta\text{Remote Share}_i, \tag{6}$$

where $l_{i,c}^{2019}$ is the employment share of industry i in county c in the pre-pandemic year 2019, and $\Delta\text{Remote Share}_i$ is the national-level change in remote work share in industry i.

In the first stage, I use this instrument to predict the county-level change in remote work share:

$$\Delta\widehat{\text{Remote Share}}_c = \alpha_1 + \beta_1 Z_c + \varepsilon_c. \tag{7}$$

In the second stage, I regress the change in the outcome variable on the predicted change in remote share:

$$\Delta Y_c = \alpha + \beta^{IV} \cdot \Delta\widehat{\text{Remote Share}}_c + \epsilon_c. \tag{8}$$

Conditional on a non-zero first stage, the validity of the instrumental variables (IV) strategy rests on three key assumptions. First, the **monotonicity** assumption requires that the instrument affects the endogenous variable, the change in remote work share, in the same direction across all counties. Second, the **exclusion restriction** stipulates that the instrument influences the outcome solely through its effect on remote work. Third, the **independence** assumption requires that the instrument is uncorrelated with unobserved determinants of the outcome. This in turn relies on the exogeneity of the pre-determined industry shares $l_{i,c}^{2019}$, which must not be systematically related to county-level unobservables that also influence changes in the outcome variable over the study period. To strengthen the case for exogeneity, I use 2019 employment shares, which are determined prior to the onset of the pandemic, and are thus plausibly uncorrelated with post-pandemic shocks or trends.

Given that the IV approach addresses both observed and unobserved selection, it is the preferred model in this study. However, in analyses where the IV approach is not appropriate—for example, owing to insufficient variation across industries—I present the DML results as the main findings.

In this study, all three approaches exploit industry-level variation in the change in remote work share and treat it as exogenous. A potential concern is that industry-specific shocks during the pandemic may be correlated with both changes in remote work share and the outcomes of interest. While the existence of such shocks cannot be directly tested, they are unlikely to threaten the validity of the results. First, the probability that industry-level shocks would systematically affect men and women differently is small. Thus, although shocks correlated with changes in remote work share may exist, they are unlikely to bias the gender-specific outcomes examined here. Second, I conduct a long-run analysis extending through the end of 2024. With a longer time horizon, the influence of pandemic-specific shocks should diminish; if the estimates were driven primarily by such shocks, the effects would be expected to weaken or disappear over time. Instead, if the effects persist, it would suggest that the results are not attributable to temporary, pandemic-specific shocks.

5 Results

This section presents the empirical findings on the impact of remote work on women's labor market share. The analysis focuses on the female share in new hires and the overall workforce. I also examine the dynamics of the effects over time. Furthermore, I explore heterogeneous effects across industries, with particular focus on those with low female representation. The goal is to assess whether remote work can serve as a tool to reduce gender segregation across industries.

5.1 The Effects of Remote Work on Female Labor Market Share

Previous research has shown that women place a higher value on job flexibility when making employment choices (Barbulescu and Bidwell, 2013; Mas and Pallais, 2017; Atkin et al., 2023). More recently, Hsu and Tambe (2025) finds that shifting a job posting from on-site to remote increases the share of female applicants by 15%. However, whether this greater willingness to apply ultimately translates into higher female representation in the workforce depends not only on supply-side preferences but also on demand-side hiring decisions, and thus remains an open question. A substantial body of work has documented that gender discrimination in hiring persists as a significant barrier in contemporary labor markets (Rivera and Tilcsik, 2016; Yavorsky, 2019; Barron et al., 2025). Building on these insights, I examine whether the rise of remote work is associated with an increase in the share of women among new hires and in the overall workforce.

Table. 1 reports estimates of the effect of changes in remote work share on female labor market outcomes, measured by the female share of new hires and the overall workforce. Columns (1) and (2) present results from the difference-in-differences (DID) specification in Equation (3) and the double machine learning (DML) model in Equation (5), respectively. Column (3) reports results from my preferred specification—the instrumental variables (IV) approach.

Table 1: The Effect of Remote Work on Hiring and Separation

	(1) β^{DD}	(2) β^{DML}	(3) β^{IV}
Panel A. Female Share in New Hires			
Δ Remote Share	0.121***	0.101***	0.097***
	(0.014)	(0.014)	(0.028)
Const.	0.509***	0.000	0.003**
	(0.000)	(0.000)	(0.001)
N	33,144	15,736	2,545
Panel B. Female Share in Workforce			
Δ Remote Share	0.040***	0.036***	0.058***
	(0.005)	(0.004)	(0.010)
Const.	0.503***	0.000***	-0.002***
	(0.000)	(0.000)	(0.000)
N	33,914	16,105	2,546

Notes: Regressions are weighted by the size of each group. Standard errors are clustered by county. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 1, Panel A presents the estimated effect of remote work on the female share of new hires across county–industry groups. Across all three specifications, the coefficient on remote work is positive and highly significant, ranging from 0.097 to 0.121. These estimates are similar in magnitude, providing consistent evidence that greater remote work adoption raises women's share among new hires. Once county fixed effects and group characteristics are controlled for, the coefficient declines slightly, from 0.121 in Column (1) to 0.101 in Column (2). My preferred specification, the IV model in Column (3), yields a coefficient of 0.097 (s.e. = 0.028). This implies that shifting all job postings in a county–industry group from in-person to remote would increase the female share of new hires by about 10 percentage points, relative to a baseline average of 49.8% in 2019.

Given the evidence that remote work has increased the female share of new hires, I next examine its effects on women's overall workforce representation. Table 1, Panel B presents the estimated impact of changes in remote work share on the change in female workforce share between 2019 and 2021. Across all three specifications, the coefficients are positive and statistically significant, ranging from 0.036 to 0.058. According to my preferred IV specification in Column (3), shifting all jobs in a county–industry group from on-site to remote is associated with a 5.8 percentage point increase in the female workforce share (s.e. = 0.010).

Relative to the baseline average of 48.2% in 2019, this magnitude may appear modest. However, placed in historical context, the effect is substantial. As shown in Figure. **??**, women's workforce share increased by only about 0.5 percentage points over the past two decades, rising from 46.5% in December 2000 to 47.0% in December 2024. Against this backdrop of slow, long-run change, the increase associated with remote work represents a dramatic acceleration in women's representation in the labor market.

Table 2: The Effect of Remote Work on Female Workforce Share, 2022 to 2025

	(1) β^{DD}	(2) β^{DML}	(3) β^{IV}
2022	0.054***	0.049***	0.070***
	(0.006)	(0.005)	(0.012)
2023	0.055***	0.049***	0.080***
	(0.006)	(0.005)	(0.016)
2024	0.056***	0.046***	0.073***
	(0.006)	(0.006)	(0.015)

Notes: Regressions are weighted by the size of each group. Standard errors are clustered by county. Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

To examine the persistence of the effects of remote work, I extend the observation window to 2022 Q4, 2023 Q4, and 2024 Q3. While the outcome variable varies across years, the treatment is fixed and defined as the change in remote job share between 2019 (pre-pandemic) and the period spanning 2020 Q4 to 2021 Q4. This design allows the estimates to capture the long-run consequences of the initial adoption of remote work on women's representation in the workforce.

The results in Table 2 indicate that the effect of remote work grows modestly over time, reaching between 7 and 8 percentage points in later years. The IV estimates are consistently larger than those from the DID and DML specifications, suggesting that the latter may be downward biased due to unobserved confounding factors. The persistence and gradual amplification of the effect have two important implications. First, they highlight remote work's potential as a lasting mechanism for improving gender representation in the labor market. Second, they provide evidence that the estimates are less likely to be driven by temporary, pandemic-specific shocks correlated with changes in remote work, and instead more plausibly reflect the causal impact of remote work itself.

In Appendix A, I also show the impact of remote work on the separation rates of males and females. Although remote work lowers the separation rate for females, it affects males to a similar extent. Therefore, I conclude that the main reason for the increase in the female workforce share is attributable to the rise in female hires.

Taken together, the evidence from both new hires and the overall workforce paints a consistent picture: remote work not only attracts more female applicants but also translates into a measurable increase in women's share in the labor market. Importantly, the workforce-level effects are smaller in magnitude than those for new hires, reflecting the gradual adjustment of workforce composition as hiring flows accumulate. Nevertheless, the fact that workforce effects remain positive and strengthen over time suggests that the initial hiring gains persist and contribute to long-run improvements in female representation in the labor market.

5.2 Heterogeneity by industry

I next examine heterogeneous treatment effects across industries, grouped by their remote work intensity and female employment share. Figure 6 in Section 3 classifies industries into four categories: (1) high-remote, female-dominated; (2) low-remote, female-dominated; (3) low-remote, male-dominated; and (4) high-remote, male-dominated. This section focuses in particular on male-dominated industries to assess whether the expansion of remote work can reduce gender-based industrial segregation.

Table 3 reports the estimated effects on the female share among new hires (Panel A) and in the overall workforce (Panel B). Across both outcomes, substantial effects are observed in male-dominated industries. According to the DML estimates,[5] if all jobs had shifted from in-person to remote in the initial post-period, the female share among new hires would have increased by 13.6 percentage points in low-remote male-dominated industries and by 3.5 percentage points in high-remote male-dominated industries. Likewise, the female workforce share in these industries would have risen by 6.1 and 1.4 percentage points, respectively.

These findings indicate that remote work is especially effective in attracting women to industries that have historically exhibited low female representation. In contrast, the effects are smaller and less consistent in female-dominated industries. The evidence highlights remote work's potential to facilitate female entry into traditionally male-dominated sectors and, in doing so, to promote a more inclusive labor market by reducing industry-level gender segregation.

Table 3: Heterogeneous Effects across Industries

	(1) High-remote Female-dominated	(2) Low-remote Female-dominated	(3) Low-remote Male-dominated	(4) High-remote Male-dominated
Panel A. Female share in new hires:				
DD	0.094	0.043*	0.149***	0.052***
	(0.100)	(0.023)	(0.021)	(0.018)
DML	0.078	0.029*	0.136***	0.035*
	(0.059)	(0.017)	(0.024)	(0.018)
Panel B. Female share in workplace:				
DD	-0.004	0.016*	0.062***	0.021***
	(0.014)	(0.009)	(0.014)	(0.006)
DML	0.008	0.020***	0.061***	0.014***
	(0.011)	(0.007)	(0.012)	(0.007)
No. of industries	1	6	7	5

Notes: Regressions are weighted by the size of each group. Standard errors are clustered by county. Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

[5] The IV approach is not appropriate in this analysis, as it is restricted to specific industries.

6 Sources of the Female Workforce Increase

The preceding analysis shows that remote work increases women's share among new hires and contributes to gradual improvements in overall workforce representation. The finding that separation rates decline by similar magnitudes for men and women further suggests that these gains are driven primarily by higher female hiring rather than lower female exits. This raises a natural next question: from where do these additional female hires originate? One possibility is that women already in the workforce switch into remote jobs through job-to-job transitions; another is that women who were previously unemployed or out of the labor force enter as new participants.

Table 4 sheds light on this mechanism by examining two potential channels of female hiring: job-to-job transitions and hires from persistent nonemployment. Due to data limitations, information on the composition of new hires is not available at the county level; therefore, the analysis is conducted at the metropolitan level. Panel A reports the estimated effect of remote work on the female share of job-to-job hires, while Panel B focuses on hires from nonemployment.

The results reveal a clear pattern. Remote work significantly increases female representation through both channels, but the magnitudes are notably larger for hires from nonemployment. According to the IV estimates, a full shift of jobs from on-site to remote would raise the female share of job-to-job hires by about 8.8 percentage points, compared with a 25.5 percentage point increase among hires from nonemployment. Because this analysis is conducted at the metropolitan level, the relatively small sample size results in wide confidence intervals, making the point estimates less precise. Nonetheless, the consistently positive and statistically significant coefficients underscore that remote work meaningfully expands women's participation through both job switching and new entry.

These findings suggest that remote work not only reshuffles women across existing jobs but also brings new entrants into the labor market. By lowering barriers to labor force participation——particularly for women facing constraints such as caregiving responsibilities——

remote work expands the pool of female workers and sustains long-run gains in women's workforce representation.

Table 4: Sources of the increase in the female share

	(1) β^{DD}	(2) β^{DML}	(3) β^{IV}
Panel A. Female Share in Job-to-Job Hires			
Δ Remote Share	0.081***	0.064***	0.088*
	(0.016)	(0.016)	(0.053)
Const.	0.505***	0.000	0.010***
	(0.000)	(0.000)	(0.003)
N	8,686	4,343	279
Panel B. Female Share in Hires from Persistent Nonemployment			
Δ Remote Share	0.141***	0.061***	0.255***
	(0.015)	(0.019)	(0.066)
Const.	0.508***	0.000	-0.011***
	(0.000)	(0.000)	(0.003)
N	8,664	4,332	279

Notes: Regressions are weighted by the size of each group. Standard errors are clustered by county. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

7 Robustness Check

The validity of the difference-in-differences (DID) framework relies on the parallel trends assumption, which requires that, absent differential treatment, groups would have followed similar trends. In the context of this study, this means that without variation in remote work adoption, the female share of new hires and the overall workforce would have evolved similarly across groups.

For the DID and DML specifications, I test this assumption by examining pre-trends. Specifically, I re-estimate Equation (3) and Equation (5) using the change in female shares from 2017 to 2019—a period prior to the sharp increase in remote work—as the outcome variable. If groups that later experienced larger increases in remote work share had already exhibited stronger gains in female representation, this would raise concerns about endogeneity. The results, reported in Table 5, show no evidence of systematic pre-trends. A small negative pre-trend of –0.008 is observed in the DID estimates for female workforce share, but the magnitude is modest. This suggests that the true effect may be even larger than the DID estimate of a 4 percentage point increase.

The identification of the IV approach depends on the exogeneity of the industries shares. The underlying assumption is that counties with different initial industries shares would have counterfactually evolved in a similar way, thus the predicted remote work change in Equation (8) is a exgenous shock. Thus, I examine the relation between the industries shares and the aggregated instrument and the pretrends of the county. I regress the change in female shares of the county on it's industries shares and the instrument. Table 6 show the results for the shares of top 5 high-remote industries and the instrument. Similarly, I don't find obvious pretrends. The industries shares of the county is not related to the trend of the female share in new hires and workforce, suggest that there is a remote shock in 2020 that causes the differences in the change of female shares.

Table 5: Relationship between change in remote shares and pre-trends in outcomes

	β^{DD}	β^{DML}
Female Share of New Hires	-0.013	-0.028
	(0.014)	(0.024)
Female Workforce Share	-0.008***	0.002
	(0.003)	(0.004)

Notes: Regressions are weighted by the size of each group. Standard errors are clustered by county. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

In this study, the validity of the IV approach hinges on the exogeneity of counties' industry composition in 2019. The key identifying assumption is that, absent differences in the change in remote work share, counties with different initial industry shares would have evolved along similar paths. Under this assumption, the predicted change in remote work share—constructed from national industry-level changes in remote work share (Equation (7))—can be interpreted as an exogenous shock. To assess this, I examine the relationship between counties' pre-2020 industry shares and pre-trends in the female share. Specifically, I regress the change in county-level female shares between 2017 and 2019 on industry shares and the instrument. Table 6 reports results for the top five high-remote industries, as well as the aggregated instrument. Consistent with the DID checks, I find no evidence of systematic pre-trends. Counties with higher initial concentrations in high-remote industries did not experience differential trends in female hiring or workforce share prior to the pandemic.

These results provide reassurance that the main findings are not driven by pretrends and support the interpretation of the estimated effects as causal.

Table 6: Relationship between county industry shares and pre-trends in outcomes

	(1) Z_c	(2) Information	(3) Professional	(4) Finance	(5) Administrative	(6) Real Estate
Female share in new hires	-0.086	0.003	-0.020	0.026	-0.030*	0.072
	(0.074)	(0.027)	(0.013)	(0.023)	(0.016)	(0.060)
Female workforce share	0.007	0.009	-0.002	0.001	0.011	0.063**
	(0.033)	(0.010)	(0.005)	(0.008)	(0.007)	(0.031)

Notes: Regressions are weighted by the size of each group. Standard errors are clustered by county. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

8 Discussions

The findings of my study have practical implications for both public policy and organizational strategy. First, as remote work increases the female share in both new hires and the overall workforce, it can serve as a policy tool to improve female representation in the labor market and promote gender equality. Importantly, this impact is not confined to traditionally female-dominated industries; rather, the effects in male-dominated industries are significant and even larger, suggesting that remote work may be a powerful tool to reduce gender segregation in the labor market. Second, my findings show that part of the increase in the female share of new hires associated with remote work comes from women who were previously persistently unemployed. This indicates that remote work encourages female labor force participation. Such an effect is particularly meaningful in the context of population aging and declining fertility rates, where policies that raise participation are critical for addressing labor shortages. Finally, the research provides a valuable perspective for firms to weigh the costs and benefits of remote work arrangements in terms of workforce stability and diversity.

This study has several limitations that also suggest directions for future research. First, the analysis focuses on the ultimate labor market outcomes of remote work and does not disentangle whether the observed increase in female representation is driven primarily by labor supply or by labor demand. While increases in female hires and employment are directly related to women's labor supply decisions, remote work may also reduce employer discrimination—for instance, by lowering employers' perceived risk that women of childbearing age will exit the labor market. Future research could distinguish between these mechanisms to better identify the channels through which remote work affects the outcomes. Second, due to data restrictions, a job is defined as "remote" as long as it includes at least one remote day. The effects of fully remote jobs may therefore be larger than the estimates presented here, and future studies with richer data could shed light on the heterogeneity between hybrid and fully remote work. Finally, the estimated effect on the female share in the overall workforce should be interpreted as a local effect. The observed changes reflect a combination of job switchers and new entrants. A comprehensive estimate at the aggregate level would require isolating the contribution of new entrants alone, which would likely yield smaller effects.

9 Conclusion

The study provides empirical evidence on the impact of remote work on female labor market representation. First, the findings show that remote work significantly increases the female share among new hires and in the overall workforce. Transitioning jobs from a fully on-site schedule to one that includes remote working days results in a 10-percentage-point increase in the female share of new hires. These gains in hiring translate into broader workforce changes, raising the overall female workforce share by about 6 percentage points. Moreover, the effect on women's workforce share is persistent over time. Second, the effects on the female share in both new hires and the workforce are particularly pronounced in male-dominated industries, suggesting that remote work has the potential to reduce gender segregation in the labor market. Third, by exploring the sources of the increase, I find that it is driven by both job switchers and new entrants.

The findings highlight that rigid job structures with fixed hours and locations continue to be major barriers to women's participation. By relaxing these constraints, remote work emerges as a powerful mechanism for advancing women's representation and reducing gender segregation in the labor market. The results carry important implications for organizations and policymakers: remote work can serve as a tool to promote diversity and equity in the workplace. Moreover, in the context of population aging and declining fertility rates, expanding flexible work arrangements may help mitigate labor shortages by mobilizing greater female labor force participation.

Future research should investigate the mechanisms behind these effects, particularly the relative roles of supply-side responses versus demand-side changes in employer behavior, and compare the impacts of hybrid versus fully remote work. Despite these open questions, the evidence presented here underscores that remote work is not only a workplace innovation but also a potential lever for promoting gender equality and sustaining labor market growth.

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Appendix A. The Effect of Remote Work on Separation

Table 7 shows the impact on separation rates separately for female and male workers. In this context, remote share of job postings is used as an proxy of remote work change of all jobs. For females, all jobs changing from in-person to at least one-day remote reduces separation rates by 12 percentage points (column (3)). This suggests that remote work helps retain female employees. The effects for males are similarly negative and significant, although slightly smaller in magnitude in the IV estimates. This parallel decline indicates that remote work benefits both genders in terms of job retention.

Table 7: The Effect of Remote Work on Separation

	(1) β^{DD}	(2) β^{DML}	(3) β^{IV}
Female			
Δ Remote Share	-0.103***	-0.0834***	-0.117***
	(0.012)	(0.010)	(0.031)
Const.	0.159***	0.000	0.024***
	(0.000)	(0.000)	(0.001)
N	32,182	15,281	2,465
Male			
Δ Remote Share	-0.101***	-0.101***	-0.080***
	(0.018)	(0.017)	(0.023)
Const.	0.166***	0.000	0.016***
	(0.000)	(0.000)	(0.001)
N	32,404	15,392	2,465

Notes: Regressions are weighted by the size of each group. Standard errors are clustered by county. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.